SAT MATH

Algebra • Practice 1.1A • Answer key

Practice 1.1A • Linear Equations in One Variable

Answer key with worked solutions

Directions

These questions follow the format of the SAT Math section. A calculator is allowed on every question (Desmos is built into the digital SAT). Multiple-choice questions have one best answer. For **student-produced responses**, write your answer on the line: a fraction or a decimal is accepted (for example $3/4$ or 0.75); give negative answers with a minus sign; do not use mixed numbers.

Question 1

What value of x is the solution to $7x - 4 = 3x + 20$?

(A) 4

(B) 6

(C) $\dfrac{16}{10}$

(D) 24

Solution

Answer: (B). $4x = 24$, so $x = 6$.

Question 2

If $2(x + 5) - 3 = 5x - 8$, what is the value of x?

(A) 5

(B) $\dfrac{5}{3}$

(C) -5

(D) 15

Solution

Answer: (A). $2x + 7 = 5x - 8 \Rightarrow 15 = 3x \Rightarrow x = 5$.

Question 3

In the equation $4(3x - 1) = 12x + c$, c is a constant. If the equation has infinitely many solutions, what is the value of c?

(A) 4

(B) -1

(C) -4

(D) 12

Solution

Answer: (C). $12x - 4 = 12x + c$ is true for every x only when $c = -4$.

Question 4

If $3x + 7 = 22$, what is the value of $6x - 1$?

(A) 5
(B) 29
(C) 30
(D) 43

Solution
Answer: (B). Double the equation: $6x + 14 = 44$, so $6x = 30$ and $6x - 1 = 29$.

Question 5

A gym charges a \$50 joining fee and \$25 per month. Awa has paid \$325 in total. Which equation gives the number of months m she has paid for, and what is m?

(A) $50m + 25 = 325$; $m = 6$
(B) $25m - 50 = 325$; $m = 15$
(C) $50 + 25m = 325$; $m = 13$
(D) $50 + 25m = 325$; $m = 11$

Solution
Answer: (D). $25m = 275$, so $m = 11$.

Question 6

What is the solution to $\frac{x}{2} + \frac{x}{5} = 14$?

(A) 20
(B) 10
(C) $\frac{98}{10}$
(D) 70

Solution
Answer: (A). Multiply by 10: $5x + 2x = 140$, $7x = 140$, $x = 20$.

Question 7

STUDENT-PRODUCED RESPONSE

What is the solution to $0.4(x - 5) = 0.3x + 1.2$?

Solution
Answer: 32. $0.4x - 2 = 0.3x + 1.2 \Rightarrow 0.1x = 3.2 \Rightarrow x = 32$.

Question 8

STUDENT-PRODUCED RESPONSE

In the equation $5(x-k)=5x-35$, k is a constant. If the equation has infinitely many solutions, what is the value of k?

Solution

Answer: 7. $5x-5k=5x-35$ for all x when $5k=35$, $k=7$.

Question 9

STUDENT-PRODUCED RESPONSE

What value of x satisfies $\frac{2x+1}{3}=\frac{x+4}{2}$?

Solution

Answer: 10. Cross-multiply: $2(2x+1)=3(x+4)\Rightarrow 4x+2=3x+12\Rightarrow x=10$.

Question 10

Which equation has no solution?

(A) $2(x+1)=2x+2$

(B) $3x+1=2x+1$

(C) $2(x+1)=2x+3$

(D) $4x=0$

Solution

Answer: (C). $2x+2=2x+3$ simplifies to $2=3$, which is false for every x. A has infinitely many solutions; B and D have $x=0$.